References

1. Pomeranz HD, Smith KH, Hart WM Jr, Egan RA. Sildenafil-associated nonartentic anterior ischemia optic neuropathy. Ophthalmology 2002;109:584–7.
2. Beck RW, Savino PJ, Repka MX, et al. Optic disc structure in anterior ischemic optic neuropathy. Ophthalmology 1984;91: 1334–7.
3. Vardi Y, Klein L, Nassar S, et al. Effects of sildenafil citrate (Viagra) on blood pressure in normotensive and hypertensive men. Urology 2002;59:747–52.
4. Mahmud A, Hennessy M, Feely J. Effect of sildenafil on blood pressure and arterial wave reflection in treated hypertensive men. J Hum Hypertens 2001;15:707–13.
5. O'Rourke MF, Nichols WW. Potential for use of pulse wave analysis in determining the interaction between sildenafil and glyceryl trinitrate. Clin Cardiol 2002;25(6):295–9.
6. Vlachopoulos C, Hirata K, O'Rourke MF. Pressure-altering agents affect central aortic pressures more than is apparent from upper limb measurements in hypertensive patients: the role of arterial wave reflections. Hypertension 2002;38:1456–60.
7. Hayreh SS. Anterior ischemic optic neuropathy: trouble waiting to happen [letter]. Ophthalmology 2000;107:407–10.

Author reply

Dear Editor:

I am in complete agreement with Dr. Gandhi's comments. One of the reasons that systemic hypotension was not mentioned as a possible mechanism for ischemia to the optic nerve in association with sildenafil use in our article was the absence of microvascular risk factors in all of the five patients described. Two of the five patients had microvascular risk factors. Patient 4 had a history of nonarteritic anterior ischemic optic neuropathy (NAION) in the fellow eye. Patient 5 was a smoker and had coronary artery disease and diabetes. The other three patients did not have any identifiable microvascular risk factors. It is certainly possible that systemic or nocturnal hypotension could have been a risk factor for some or all of these patients.

Since this article was published, 10 additional cases of sildenafil-associated NAION have come to my attention. A summary of these cases is now being prepared for submission for publication. Interestingly, all of these additional 10 patients have microvascular risk factors, including hypertension, diabetes, or hyperlipidemia. Three of the patients had bilateral sequential NAION. One of them had final visual acuities of light perception (right eye) and hand motions (left eye).

I thank Dr. Gandhi for bringing the readers' attention to this important possible mechanism for sildenafil-associated NAION.

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Latanoprost versus Bimatoprost

Dear Editor:

Gandolfi and Cimino[1] report on a small series of patients who did not respond to latanoprost but did respond to bimatoprost. I congratulate them on their planned, prospective, balanced, crossover design. This design is very powerful for discerning small differences by diminishing the effects of biologic variability.

There are several items reported that need clarification. The characterization of bimatoprost as a prostamide is speculative and without literature support. The reference cited by the authors is a theoretical construct article without data.[2] Repetition of a theory does not produce a more valid statement. There is evidence that the bimatoprost free acid activates the human $F_{2\alpha}$ receptor, similar to latanoprost and travoprost.[3] There is also evidence that bimatoprost amide (Lumigan, Allergan, Inc., Irvine, CA) is hydrolyzed to the free acid by the human cornea.[4] Therefore it is more likely that bimatoprost is a prostaglandin analogue, the same as latanoprost and travoprost, but with poor corneal penetration. This would explain the need for the sixfold higher concentration of bimatoprost (0.03%) compared with latanoprost (0.005%). It is possible that some eyes are deficient in corneal esterase ability and may not be able to convert latanoprost into the free acid adequately. This is a more plausible explanation for the study findings than the prostamide theory.

The authors cite a prior work purporting to show "a greater percentage of eyes reaching target [intraocular pressures] in the low teens,"[5] yet have ignored published criticism regarding the incomplete and incorrect method of analysis to achieve this result.[6] The cited study showed no difference in intraocular pressure (IOP)–lowering ability between bimatoprost and latanoprost over the 3-month period. It is not valid to extract a subset of data for special analysis as was done for the target IOP results.

Lastly, it is difficult to generalize the findings of this study to clinical practice due to the highly selected study group. Everyone who practices medicine encounters patients who respond in unusual ways. Clinically there are patients who respond to latanoprost but do not respond to bimatoprost. What would a study of this group prove? The missing data from the present study are the denominator. How many patients were screened, in total, to find the group studied? Only by knowing the probability of finding such a patient can we make sense of the findings.

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References

1. Gandolfi SA, Cimino L. Effect of bimatoprost on patients with primary open-angle glaucoma or ocular hypertension who are nonresponders to latanoprost. Ophthalmology 2003;110:609–14.
2. Woodward DF, Krauss AH, Chen J, et al. The pharmacology of bimatoprost (Lumigan). Surv Ophthalmol 2001;45(Suppl 4): S337–45.
3. Sharif NA, Kelly CR, Crider JY. Agonist activity of bimatoprost, travoprost, latanoprost, unoprostone isopropyl ester and other prostaglandin analogs at the cloned human ciliary body FP prostaglandin receptor. J Ocul Pharmacol Ther 2002;18: 313–24.
4. Maxey KM, Johnson JL, LaBrecque J. The hydrolysis of bimatoprost in corneal tissue generates a potent prostanoid FP receptor agonist. Surv Ophthalmol 2002;47(Suppl 1):S34–40.
5. Gandolfi S, Simmons ST, Sturm R, et al. Three-month com-